Labor Market Sorting over the Business Cycle

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Abstract

I study whether workers selectively separate from their job matches in a recession. Using data from the Current Population Survey, I show that workers with short tenure and high wages are more likely to lose their jobs and become unemployed in recession periods. Moreover, the weekly history of workers from National Longitudinal Surveys suggests that workers having an experience of separation into unemployment for more than four weeks have larger separation probabilities later in their career. Moreover, the gap in separation probabilities between workers with and without separation experience becomes wider in recessions, which suggests an abundance of workers with small expected match surplus, or high separation rates, in the unemployment pool in bad times. I then test whether selective separation at the onset of the recession affects the job creation decision of employers. Since vacant jobs are more likely to be matched with unemployed workers, cyclical variation in the composition of unemployed workers changes the expected return to post vacancies. I introduce workers heterogeneous of outside values into the Diamond-Mortensen-Pissarides model with endogenous separation to see the macroeconomic implications of worker sorting through employment status over the business cycle. The key mechanism is that the firms tend to separate from their low surplus employees more frequently during recessions, entrants take this account, and they reduce the number of vacancies created. Although the composition change amplifies cyclical fluctuations in job-finding rates and unemployment rates, its quantitative importance is small, compared to underlying forces to create cyclical fluctuations. I plan to quantitatively evaluate the role of sorting in generating the cyclical cost of job loss.

Keywords: cyclical unemployment, endogenous separation, labor market sorting, tenure dependence of separation probabilities, worker heterogeneity.

JEL codes: J63, E24, J64

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1 Motivation

A typical response of the labor market to a recession is characterized by a sharp increase in workers' separation probabilities at the onset of the recession and more persistent reduction in job-finding probabilities of unemployed workers. If cyclical changes in labor market flows are different across *ex ante* heterogeneous workers, potential employers might adjust their job creation behavior to the composition of unemployed workers in a recession. In this project, I study how cyclical variation in the composition of unemployed workers is important in the job creation process over the business cycle. I argue that the fraction of unemployed workers that are low type increases in recession periods. Since entering firms have higher probabilities of matches to unemployed workers, this cyclical change in worker composition lowers the entry value.

To support this idea empirically, I show the unemployed worker pool has more inflows of high wage and short tenure workers in recessions, using the Current Population Survey (henceforth, CPS) micro-data. Workers in the lowest 33 percent wage group in an occupation and year cell have monthly job losing probabilities higher by 0.15 percent than the top 33 percent wage group. In recessions, the gap between wage groups disappears, implying an average unemployed worker has higher wage in her previous job.

On the other hand, workers in the shortest 33 percent tenure group are 0.75 percent more likely to become job losers within a month, and the difference widens to 1 percent in recession periods. If the short tenure of workers implies low match surplus, and this component stays persistent over worker career, having more short tenure in the match pool indicates having higher probabilities of small match surplus to the firms. I use the long panel of workers from National Longitudinal Surveys of Youth 1979 (hence, NLSY79) to test the persistence of low match surplus component over workers' labor market transition. Having an experience of transition from employment to unemployment (EU transition) longer than four weeks increases average weekly EU transition rates by 0.2 percent, compared to workers without such experiences. During recessions, this effect amounts to 0.4 percent.

Data suggests that there exists a worker group with relatively higher wage and shorter

tenure, and this worker group is more vulnerable to cyclical employment risk than others. A typical example of this kind of workers is lemons. When screening and signaling are imperfect in the hiring process, the firms that end up hiring low type workers frequently separate from these workers. The firms are more likely to do so in recessions since the revenue of the match declines. If the wage cannot adjust freely to the decline in match productivity, the match ends with higher probabilities.

Another example of the worker group with short tenure and high wage is low productivity workers working in high paying firms. If cross-sectional wage distribution within a firm is restricted so that the wage cannot reflect the productivity of workers entirely, the firms separate from those workers more frequently, and the selection over time implies that the short-tenured worker is more likely to get fired. Recent studies using employer-employee matched data suggest that high paying firms and low productivity workers separate disproportionately more during recessions.

I construct a simple Diamond-Mortensen-Pissarides (hence, DMP) type model to account for the cross-sectional difference in cyclical fluctuations in separation probabilities. Workers are heterogeneous in terms of the disutility of work, and their productivity. Low match surplus workers have high disutility of work and low productivity, and high match surplus ones have low disutility of work and high productivity. Endogenous separation happens when an overhead cost shock hits the existing match, and the realized cost is higher than the match surplus. As a result, at a given wage, only low match surplus workers separate more frequently. Because of selection over the tenure, short labor market tenure is associated with higher probabilities of being low type and higher separation probabilities. Match-specific productivity implies a positive(negative) correlation between wage and the match surplus(separation probabilities).

In a recession, low type workers with low match-specific productivities experience the largest increase in separation probabilities. These workers have a higher wage because of higher disutility from work, and they are more likely to be short-tenured, which replicates

the data pattern observed from the CPS and the NLSY79.

The selective separation process means that the pool of unemployed workers changes over the business cycle. As long as the match is random across the type of workers, the employer is more likely to match with low type workers and have less match surplus. Thus, the free entry condition implies that the labor market tightness decreases because of the composition effect. To quantify the importance of this margin, I compare the baseline impulse-response function to the counterfactual impulse-response function without the compositional effect in the free entry condition. The labor market sorting over the business cycle implies larger and more persistent dynamics in job-finding probabilities and unemployment rates, but it has limited quantitative importance.

2 Literature Review

This paper is in line with recent literature on unobservable characteristics of workers, and their effects on labor market flows. Using CPS microdata to observe short career history of workers, Hall and Kudlyak (2019) assume five types of worker groups with different transition rates and try to decompose labor market flow rates as a mixture of heterogeneous exponential processes across long-term jobs, short-term jobs, non-participation, and unemployment. Morchio (2016) analyzes NLSY79 and shows that prime-age unemployment is concentrated for a segment of workers and unemployment spell and separation history among young workers is a strong predictor of staying unemployed and frequent separations at the prime age. He argues that information frictions on worker type and learning over the worker career are essential in explaining these facts. This paper introduces a cyclical component into this literature and shows that the workers with separation history are more vulnerable to cyclical shocks.

While recent empirical studies emphasize heterogeneities both in job-finding and separation probabilities, equilibrium unemployment models usually endogenize only hiring

process so that the former component depends on the worker type. These papers often base themselves on the empirical evidence from experimental episodes on duration dependence of callback rates (Oberholzer-Gee (2008), Kroft et al. (2013), Eriksson and Rooth (2014)); long-term unemployed are less likely to pass the initial process of hiring. When worker heterogeneity can be observed even imperfectly in the hiring process, longer unemployment duration signals the workers being low type (Doppelt (2016), Jarosch and Pilossoph (2019)). This paper, on the other hand, argues that the short job tenure signals the workers being low type and that the selective separation pattern in recession implies the pool of unemployed workers change in a less profitable direction than what a purely random separation suggests.

Gibbons and Katz (1991) and Nakamura (2008) incorporate selection in separation into hiring process. When worker type is unobservable, but worker history is observable, employers take hiring decision based on the separation history of the worker. This paper assumes random matching, so the heterogeneity is not in the hiring process but the separation process. As a result, conditional on entry, it is assumed that the employer does not differentiate its hiring decision depending on the career history of the worker, but *before the entry*, the free entry condition takes the composition of unemployed worker into account. This assumption simplifies the state variable of an individual worker and makes the model tractable. Moreover, as Nakamura (2008) and Kroft et al. (2013) show[1], the signaling of bad employment history has weeker effect during recessions. The random matching is not a too restrictive assumption in a slack labor market this paper is interested in.

Mueller (2017) also takes cyclical variation in the pool of unemployed workers into account and explores the impact of such change on job creation incentives by firms. He argues the unemployed worker pool in the CPS exhibits a larger fraction of workers with high wages

[1]Biewen and Steffes (2010) tells there is no evidence that the negative impact of unemployment on job-finding is weaker in recessions, using German data. Since whether duration dependence in unemployment comes from stigma effect or human capital depreciation is not a subject of this paper, I rather take a neutral view and assume there is no selection in hiring process at all.

from their previous jobs during recessions, and this comes from the high cyclicality of separation rates of those workers but not from job-finding sides. However, he concludes that this compositional change dampens cyclical fluctuations in job-finding and unemployment rates, as workers with higher wage in their last job are more productive, and potential employers benefit from having higher probabilities of being matched to them. This paper argues the opposite that the workers that additionally comes into unemployment during recessions are more likely to give a lower value to the potential employers, despite their higher wage in previous jobs, and the compositional effect leads to the larger cyclicality of job-finding and unemployment rates.

Lise and Robin (2017) is one of a few studies that incorporate two-sided heterogeneity between workers and firms into the search-matching model with business cycle shocks. It builds a tractable framework to analyze the evolution of worker-firm match distribution over the business cycle. The key prediction of the model is that low type workers are fired, especially by high productivity firms when the economy enters a recession. This paper delivers a similar phenomenon where a recession is characterized by workers paid with high wage by reasons other than their productivity are more likely to flow into the unemployment in bad times. The implication of such separation processes is lower job-finding rates appearing in the free entry condition and more sluggish recovery in the labor market after a negative shock.

3 Separation Probabilities over the Business Cycle

3.1 Cyclical Pattern of Separation Probabilities over Wage and Job Tenure Groups

Is there any systematic difference across workers in cyclical fluctuations in probabilities of losing a job? If so, can it reinforce or dampen the cross-sectional importance of economic shock on the job-finding rate for a prolonged period? To address these questions, I

estimate firing probabilities over wage and job tenure bins and test how cyclical firing risk accrues differently to each bin.

For the wage bin, I use the Annual Social and Economic Supplement (ASEC) of the CPS. Then, using the short panel structure of the CPS, I merge ASEC samples with the CPS core samples in the next month and estimate the probability that someone gets fired between ASEC survey in March and the CPS core survey in April. The dependent variable $f_{i,t}$, where i is worker and t is year, is defined as

$$f_{i,t} = \begin{cases} 1 & \text{if worker i is employed in ASEC sample} \\ & \text{and becomes job loser in the next month survey} \\ 0 & \text{if worker i is employed in ASEC sample} \\ & \text{and remains employed in the next month survey} \end{cases}$$

Then, I construct three wage bins, where each worker is assigned by their rank in each year and occupation cell, after excluding the top 5 and the bottom 5 percent of wage earners from my samples. I regress the following linear probability model.

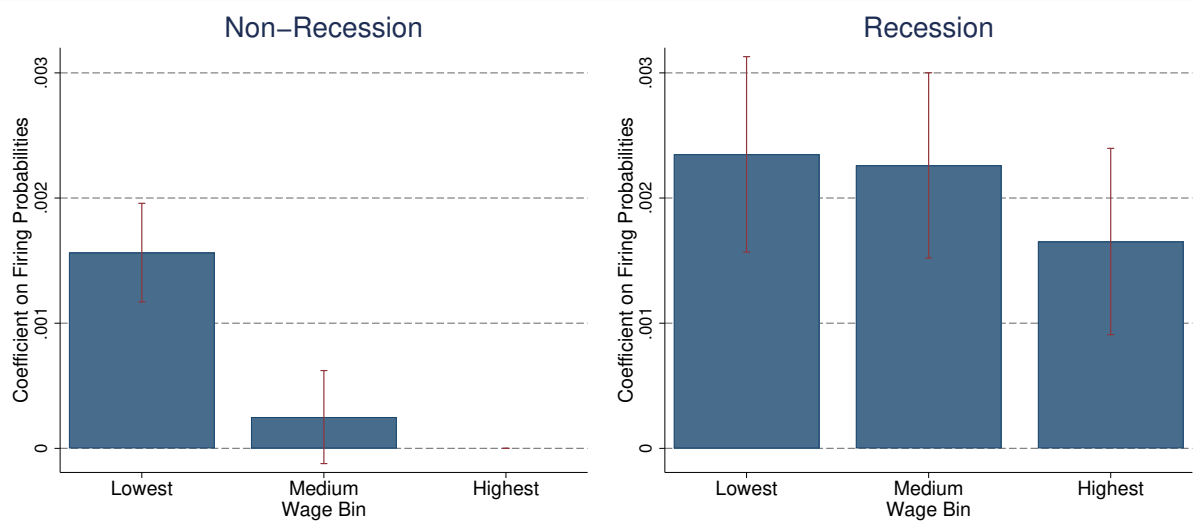
$$f_{i,t} = X'_{i,t}\beta + \sum_{q=1}^{3} \alpha_{q,normal} \mathbf{1}_{i,t}^{q,normal} + \sum_{q=1}^{3} \alpha_{q,recess} \mathbf{1}_{i,t}^{q,recess} + \nu_{i,t}$$

where $\mathbf{1}_{i,t}^{q,normal}$ is a dummy that is 1 if the worker i belongs to the wage bin q and time t is non-recession periods. Likewise, $\mathbf{1}_{i,t}^{q,recess}$ is 1 if the worker i belongs to the wage bin q and the time t is in recessions. $X_{i,t}$ includes age, squared age, and dummies on race, occupation, industry, gender, and education level.

In order to estimate job losing probabilities over job tenure bins, I use Job Tenure and Occupational Mobility Supplement (JTOM) of the CPS. This supplement is surveyed in either January or February every year. The firing dummy is defined similarly as before,

and the tenure bin is constructed by using workers' tenure rank in each occupation cell, regardless of the year. Note that I excluded workers with job tenure shorter than 0.1 years and longer than 20 years. The controls are the same as before. Figure 3 and 4 show the coefficient estimates. Notice I normalize coefficients on the highest wage bin and the longest job tenure bin in the non-recession periods to zero.

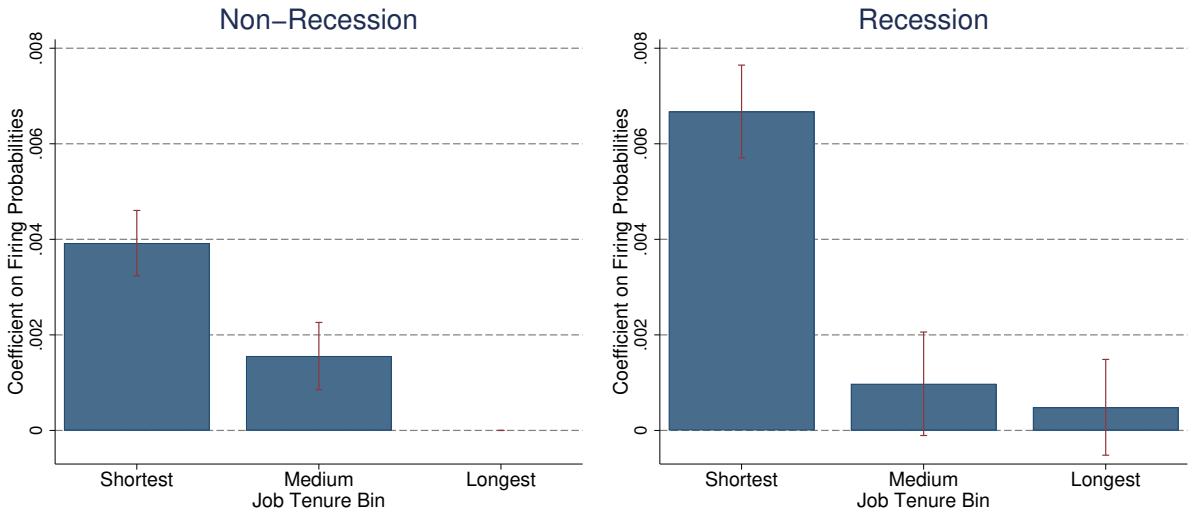
Figure 1: 1-Month Job Losing Probabilities over Wage Groups



Data Source: IPUMS-CPS, Annual Social and Economic Supplement, $N = 707,273$, $R^2 = 0.0075$,
The probability is normalized to the highest wage group in non-recession periods

In non-recession periods, the wage earners in the first wage bin are 0.15 percent more likely to become job losers within a month, compared to the ones in the third wage bin. However, this difference is insignificant during a recession, and the highest wage group have a larger increase in separation rates than the lowest wage group, which is significant at the 10 percent level. The tendency is robust to controlling for a linear trend, the metropolitan status, the part-time status of each worker, and including job leavers, but the difference is not significant under some specifications. This result is consistent with recent findings by Mueller (2017). He shows that workers with higher wage have larger cyclicality of separation rates, making them account for a disproportionately larger

Figure 2: 1-Month Job Losing Probabilities over Tenure Groups



Data Source: IPUMS-CPS, Annual Social and Economic Supplement, $N = 375,031$, $R^2 = 0.0126$,
The probability is normalized to the highest wage group in non-recession periods

fraction of unemployed worker pool in recession periods.

On the other hand, if workers are classified by their job tenure, workers in the first tenure bin have a 0.75 percent higher probability of becoming a job loser within a month. Moreover, it is the workers with the shortest job tenures that experience the largest increase in the firing probabilities. Only this group has an increase in job losing probabilities statistically significant at the 10 percent level, and the difference in the cyclical increase is also significant at the 10 percent level. The tendency and the significance of the gap between cyclical increments are robust to controlling for a linear trend, the metropolitan status, the part-time status of each worker, and including job leavers.

This exercise gives several takeaways. First, workers with lower wage and shorter job tenure are overall more likely to become job losers, and short tenure is more strongly associated with higher separation probabilities than low wage as shown from the magnitude of the estimated coefficients. Second, workers with higher wage and short tenure have distinctively larger separation probabilities during recessions than in normal times. If the

short tenure indicates worker component associated with a smaller profit to firms, despite the higher wage, differential EU rates over tenure bins imply lower entry value from the compositional change among unemployed workers.

3.2 Persistent Impacts of Separation Experience on Workers

Even if separation is a selective process to sort bad matches, it does not lead to composition changes in the pool of unemployed workers if the heterogeneities do not persist over time. This section tries to see if workers have different transition probabilities later, depending on whether they have any history of separation into unemployed or not[2]. I use worker-level panel data from NLSY79. The dataset records a weekly history of each worker's employment status that started in 1979. I test if the workers having EU transition before are more likely to separate from their jobs into unemployment or later, and if this pattern intensifies during recessions.

The EU separation variable is defined as follows. The worker is considered to separate from work only when the subsequent unemployment spell is longer than four weeks.

$$\tilde{f}_{i,t} = \begin{cases} 1 & \text{if worker } i \text{ is employed in week } t \\ & \text{and is unemployed from week } t+1 \text{ to } t+4 \\ 0 & \text{if worker } i \text{ is employed in week } t \\ & \text{and remains employed in any week from } t+1 \text{ to } t+4. \end{cases}$$

The four-week condition is to capture the idea that the worker may be involved in the job to job transition and have a short unemployment spell in between two jobs. However, most of the results are robust to removing the four-week condition and use only the employment status of one week after. By these definitions, the EN transitions are excluded from my samples. I also excluded samples that are not working but not distinguishable between

[2]The workers without separation history are re-entrants without any EU transition before and entrants.

non-participation and unemployment.

The main regressor, EU separation experience dummy, is

$$\tilde{g}_{i,t} = \begin{cases} 1 & \text{if } max\{\tilde{f}_{i,s}\}_{s<t} = 1 \\ 0 & \text{otherwise.} \end{cases}$$

Note that the worker has $g_{i,t} = 0$ if the worker separates for the first time in her life at t. The regression specification is given as

$$\tilde{f}_{i,t} = X'_{i,t}\beta + \alpha_0 \tilde{g}_{i,t} + \alpha_1 \tilde{g}_{i,t} \mathbf{1}_t^{recess} + \nu_{i,t}$$

where $X_{i,t}$ contains controls including age, squared age, and dummies on education level, sex, occupation, industry, race, and recession.. Note that then α_1 can be interpreted as the additional effect of the recession on separation probabilities of workers with separation experience. Table 1 summarizes the estimation results.

Having separation experience strongly predicts higher EU transition rates later, particularly for young workers. This finding is in line with the empirical findings in Morchio (2016). He shows that the separation experience of young workers leads to higher separation probabilities at their prime age. For the young worker samples, a worker having a separation experience before has weekly separation probability higher by 0.25 percent than workers without any EU transition experience.

This tendency is even stronger in recession; having a separation experience is associated with 0.31 percent total increase in separation probability in bad times, compared to only 0.05 percent increase for an average worker without such an experience. The estimates suggest the importance of separation experience on young workers relative to old workers in recession periods. The coefficient on the cyclical component is smaller and less significant for full samples in NLSY79 as in column (2). When the regression includes hourly

Table 1: NLSY79 Regression Estimates

Dependent Variable	Separation Probability (in %)		
	(1)	(2)	(3)
Separation Experience	0.246***	0.171***	0.245***
	(0.007)	(0.004)	(0.007)
Recession	0.047***	0.039***	0.046***
	(0.009)	(0.006)	(0.009)
Recession ×	0.264***	0.004	-0.024
Separation Experience	(0.026)	(0.013)	(0.053)
Hourly Wage			0.005***
			(0.000)
Hourly Wage × Recession ×			0.025***
Separation Experience			(0.004)
N	2640113	4902911	2231192
R-squared	0.001	0.001	0.001

Notes: ***,**,*: significant at 1, 5, 10 percent, respectively. Column (1) and (3) use young (age < 35) worker samples in NLSY79. Column (2) uses all age samples from 25 to 60. Note that I use samples from 1998 because the occupation and industry codes are not collected before then.

wage, and the additional effect of hourly wage on workers with separation experience, the level effect loses magnitude and significance, as shown in column (3).

To summarize, the workers having a separation experience behave significantly differently later in the labor market. They are more likely to separate from their jobs, and cyclical shocks make them separate particularly more than workers without such experiences. As a result of increases in job separation probabilities heterogeneous across workers, there are more unemployed workers with multiple separation experiences in downturns, which the job creation side takes into account.

To see the implication of the worker sorting over the business cycles, I construct a frictional labor market model with worker heterogeneity and endogenous separation. The goal of the model is to generate a group of workers with high wage and short job tenure to account for the cyclical patterns in firing probabilities, and the model achieves this by introducing a component of worker type, disutility from working, increasing wage but

decreasing firm profit. The model delivers qualitatively similar results on the cyclical pattern on separation probabilities. Then, I compare the baseline impulse-response functions to a counterfactual one without worker sorting.

4 Model

I build a model to account for the cross-sectional patterns in separation probabilities; a worker with higher wage and shorter tenure is associated with the largest cyclical increase in separation probabilities. If the wage reflects a larger match surplus, part of which the firm perceives as profit, the overall tenure is longer for these workers through selection, contrary to the data. Consequently, there has to be another wage-increasing factor that deteriorates the match surplus. As described above, such workers can be thought of as lemons in the labor market or low type workers working for high productivity firms. Firms happen to hire low productivity workers after an imperfect screening process and tend to separate from them more frequently. Alternatively, firms subject to wage equalization constraint, to some extent, choose to fire workers with the lowest productivity.

To generate a group of workers with high wage and small match surplus, I assume workers with low productivity have high disutility of work. A high wage might result from both high disutility of work, in case of low surplus workers, and high productivity, in case of the high type. At a given wage, the separation is concentrated around the low type workers. Through selection, short tenure workers are more likely to be low type, and the separation probability is higher for short tenure workers. On the other hand, the cyclical shock hits mostly low type workers with a higher wage, because the decrease in match surplus has a larger marginal effect on separation probabilities with a higher density of realized overhead cost shock.

The model builds on the standard Diamond-Mortensen-Pissarides model (Pissarides (2000), Chapter 1 and 6) and additionally introduces worker heterogeneity and match-specific productivity. The economy consists of risk-neutral workers and firms. Each worker has

discrete type $\theta \sim m(\theta)$. Note that the type affects both productivity and the disutility of work. One firm matches to one worker, after posting vacancies that incur costs to them. Once a firm matches to a worker, they draw match-specific productivity ξ from distribution $H(\xi)$, which is time-invariant.

Each worker-firm match encounters overhead cost shock with a Poisson rate of Ψ, which is potentially time-varying. Conditional on the shock, the overhead cost is distributed as continuous distribution $F(c)$ with full support over $\mathfrak{R}^+$. In order to continue the production, the match has to pay the overhead cost. In equilibrium with the possibility to negotiate in case of the match continues only if the realized cost is smaller than the match surplus, defined as the additional value of the production relative to the sum of outside values.

To put this piece into the standard real business cycle framework, I assume that there is a representative household. It gathers all labor income together and decides how much to consume and to save. Their savings are used to produce by the firm-worker matches or to create vacancies. All the profits are pooled together and paid to the households, net of new vacancy costs. Assume an individual worker-firm match has the following decreasing return to scale production technology.

$$y = \xi \theta k^{\alpha},$$

where k is the capital input of individual match, with the rental rate r, ξ is firm-side productivity, and θ denotes worker type.

After maximization, a typical match earns the following revenue per period.

$$\pi(\theta, \xi) = (\theta\xi)^{\frac{1}{1-\alpha}} r^{\frac{-\alpha}{1-\alpha}} (\alpha^{\alpha} - \alpha^{\frac{1}{1-\alpha}}).$$

4.1 Workers

Workers are risk-neutral. Employed workers receive wage and separate from their jobs exogenously and endogenously. Working incurs disutility of work $b(\theta)$ to them, and the disutility is type-specific. The following equation describes the values of workers with type θ, match specific-productivity ξ, and wage w.

$$\begin{aligned} r\bar{E}(\theta,\xi,w|\Omega) = w - b(\theta) &+ (\chi + \Psi\mathbb{P}\left[S(\theta,\xi|\Omega) < c\right])(U(\theta|\Omega) - \bar{E}(\theta,\xi,w|\Omega)) \\ &- \Psi\mathbb{E}[X(c)|S(\theta,\xi|\Omega] \geq c]\mathbb{P}\left[S(\theta,\xi|\Omega) \geq c\right] \\ &+ \frac{d}{dt}\mathbb{E}[\bar{E}(\theta,\xi,w|\Omega)|\Omega], \end{aligned}$$

where $S(\theta,\xi|\Omega) := \bar{V}(\theta,\xi,w|\Omega) + \bar{E}(\theta,\xi,w|\Omega) - U(\theta|\Omega)$ is the match surplus function. Ω is the aggregate state variable, composed of the frequency of cost shock Ψ, common across agents, and the distribution of workers $g(\xi,\theta)$. I assume the agents can bargain how to split the overhead cost freely, so the match ends only when the cost exceeds the match surplus. Consequently, the match ends with exogenous rate ξ and endogenous rate $\Psi\mathbb{P}\left[S(\theta,\xi|\Omega) < c\right]$, the probability that the overhead cost is larger than the match surplus multiplied by the Poisson rate. The second line indicates the expected overhead cost the worker has to pay when the total cost is c. It is assumed that the amount the worker pays to cover the overhead cost depends only on the realized cost of c. Note that $X(c) \leq \bar{E}(\theta,\xi,w|\Omega)$ in equilibrium. The wage us pinned down below through Nash-bargaining.

When the worker is unemployed, the value can be expressed as

$$rU(\theta|\Omega) = p(\tau)\int \left(E(\theta,\xi|\Omega) - U(\theta|\Omega)\right) dH(\xi) + \frac{d}{dt}\mathbb{E}[U(\theta|\Omega)|\Omega],$$

where $p(\tau)$ is the rate at which each worker is matched to a firm, which is determined in equilibrium by the market tightness $\tau = v/u$. Note that the flow utility of unemployed

workers is normalized to zero.

4.2 Firms

The incumbent firm earns $\pi(\theta, \xi)$ from production every period and pays a wage to the worker. The firm value function is given as

$$\begin{aligned} r\bar{V}(\theta, \xi, w|\Omega) = \pi(\theta, \xi) - w + (\chi + \Psi\mathbb{P}\left[S(\theta, \xi|\Omega) < c\right])(-\bar{V}(\theta, \xi, w|\Omega)) \\ - \Psi\mathbb{E}[c - X(c)|S(\theta, \xi|\Omega) \geq c]\mathbb{P}\left[S(\theta, \xi|\Omega) \geq c\right] \\ + \frac{d}{dt}\mathbb{E}[\bar{V}(\theta, \xi, w|\Omega)|\Omega]. \end{aligned}$$

The firm pays the rest of the overhead cost after the worker pays a part of it. In equilibrium $c - X(c) \leq \bar{V}(\theta, \xi, w|\Omega)$. Note that the outside value for the firm is normalized as one.

4.3 Wage and Match Surplus

The wage is pinned down through Nash-bargaining. To be specific, the worker and the employer bargain over wage to maximize the following object, with firm share parameter η.

$$w = \arg\max_{w} \left(\bar{E}(\theta, \xi, w|\Omega) - U(\theta|\Omega)\right)^{1-\eta} \left(\bar{V}(\theta, \xi, w|\Omega)\right)^{\eta}$$

Assume Nash bargaining with the same parameter η on the overhead cost.

$$\begin{aligned} X(c) &= (1 - \eta)c \\ c - X(c) &= \eta c \end{aligned}$$

Denote the resulting wage $w(\theta, \xi|\Omega)$ and define $V(\theta, \xi|\Omega) := \bar{V}(\theta, \xi, w(\theta, \xi|\Omega)|\Omega)$ and $E(\theta, \xi|\Omega) := \bar{E}(\theta, \xi, w(\theta, \xi|\Omega)|\Omega)$. Then, the first order condition of the problem above implies

$$E(\theta, \xi|\Omega) - U(\theta|\Omega) = (1-\eta)S(\theta, \xi|\Omega)$$
$$V(\theta, \xi|\Omega) = \eta S(\theta, \xi|\Omega).$$

The function $S(\theta, \xi|\Omega)$ satisfies

$$\begin{aligned} rS(\theta, \xi|\Omega) = \pi(\xi|\Omega) &+ \big(\chi + \Psi\mathbb{P}[S(\theta, \xi|\Omega) < c]\big)\big(-S(\theta, \xi|\Omega)\big) \\ &- b(\theta) - p(\tau)\int \Big(E(\theta, \tilde{\xi}|\Omega) - U(\theta|\Omega)\Big)\, dH(\tilde{\xi}) \\ &- \Psi\mathbb{E}[c|S(\theta, \xi|\Omega) \geq c]\mathbb{P}[S(\theta, \xi|\Omega) \geq c] \\ &+ \frac{d}{dt}\mathbb{E}[S(\theta, \xi|\Omega)|\Omega]. \end{aligned}$$

As a result, it is sufficient to keep track of only $S(\theta, \xi|\Omega)$, not $E(\theta, \xi|\Omega)$ and $V(\theta, \xi|\Omega)$ separately. Rearranging the equation above gives the following formulation for the negotiated wage.

$$\begin{aligned} w(\theta, \xi) &= (1-\eta)\pi(\theta, \xi) + \eta b(\theta) + \eta p(\tau)\int \Big(E(\theta, \tilde{\xi}|\Omega) - U(\theta|\Omega)\Big)\, dH(\tilde{\xi}) \\ &= (1-\eta)\pi(\theta, \xi) + \eta b(\theta) + \eta(1-\eta)p(\tau)\int S(\theta, \xi|\Omega) dH(\tilde{\xi}). \end{aligned}$$

In the exercise below, the observed wage and labor share have to take this overhead cost into account. Firms pay the overhead cost, and this cost is considered as the non-labor share of the production. Moreover, the matches with the same (θ, ξ) are insured against this overhead shock, and each match pays the expected overhead cost lower than their match surplus every period. The worker pays η fraction of the expected cost. Then the

problem stays the same, but the observed wage becomes

$$\tilde{w}(\theta,\xi) = (1-\eta)\pi(\theta,\xi) + \eta b(\theta) + \eta(1-\eta)p(\tau)\int S(\theta,\xi|\Omega)dH(\tilde{\xi})$$
$$- \Psi\mathbb{E}[c|S(\theta,\xi|\Omega) \geq c]\mathbb{P}[S(\theta,\xi|\Omega) \geq c].$$

where the last term is the insurance fee.

From now on, I denote the endogenous separation rate as $\delta(S(\xi,\theta|\Omega)) = \Psi\mathbb{P}[S(\xi,\theta|\Omega) < c]$ and the expected cost the match chooses to pay conditional on the cost shock as $\mathbb{C}[S(\theta,\xi|\Omega)] = \mathbb{E}[X(c)|S(\theta,\xi|\Omega] \geq c]\mathbb{P}\left[S(\theta,\xi|\Omega) \geq c\right]$ for notational simplicity.

4.4 Matching Technology, Free entry, and the Market Clearing

The matching between firms and workers follows the Cobb-Douglas technology;

$$m(u,v) = Qv^{\gamma}u^{1-\gamma},$$

which gives $p(\tau) = Q\tau^{\gamma}$ and τ=v/u.

The free entry condition is given as

$$c = q(\tau)\int\sum_{\theta}\frac{u(\theta)}{u}\eta S(\theta,\xi|\Omega)dH(\xi)$$

The distribution is characterized by

$$\dot{g}(\theta,\xi) = u(\theta)p(\tau)h(\xi) - (\chi + \delta(S(\theta,\xi|\Omega)))g(\theta,\xi)$$

The representative household gathers all income from wage and firm profit. It has log

utility from consumption and discount rate ρ. The representative household decides how much to save and consume given economy-wide interest rate r^C. Then, the standard Euler equation characterizes the optimal policy of the household.

$$\frac{d}{dt}\mathbb{E}\left[\frac{1}{c(\Omega)}\right] = (\rho - r^C(\Omega))\frac{1}{c(\Omega)},$$

which gives the interest rate as $r^C = \rho$ and the rental rate of capital as $r = \rho + \delta$ in the steady state.

Lastly, final good and the capital market clearing conditions are given as

$$\begin{aligned}
\frac{d}{dt}K(\Omega) + c(\Omega) + c_v v(\Omega) &= \left(\frac{\alpha}{r(\Omega)}\right)^{\frac{\alpha}{1-\alpha}} \sum_{\theta} \int (\xi\theta)^{\frac{1}{1-\alpha}} g(\theta,\xi) d\xi \\
&\quad - \sum_{\theta} \Psi \int \mathbb{C}(S(\theta,\xi|\Omega)) g(\xi,\theta) d\xi - \delta K(\Omega) \\
K(\Omega) &= \left(\frac{\alpha}{r(\Omega)}\right)^{\frac{1}{1-\alpha}} \sum_{\theta} \int (\xi\theta)^{\frac{1}{1-\alpha}} g(\theta,\xi) d\xi
\end{aligned}$$

By Walras' law, the market clearing conditions ensures that the budget constraint of the representative housed holds with equality in equilibrium.

4.5 Equilibrium under the Aggregate Uncertainty

When the aggregate fluctuations come from stochastic process to the aggregate process, the fluctuations in the labor market are too small, as well-known in the literature. To amplify cyclical responses of the labor market, suppose the aggregate uncertainty comes from exogenous shocks to the overhead cost frequency Ψ, which follows the Ornstein-

Uhlenbeck process.

$$d(\Psi_t - \Psi) = -\beta(\Psi_t - \Psi)dt + \sigma dW_t$$

where W_t is the Wiener process, β mean-reversion parameter, σ variance of the driving force. A positive shock to Ψ_t increases separation probabilities through more frequent overhead cost shock and smaller match surplus from an increase in expected overhead cost. Entry is also reduced by the decrease in expected match surplus.

Following Reiter (2010) and Ahn et al. (2018), I discretize the space for ξ and linearize[3] the system of equations that characterize the equilibrium path with aggregate uncertainty around the deterministic steady state and apply the Blanchard-Kahn to pin down the forward-looking variables, such as the value functions and the consumption level. The system of equations is fully characterized in appendix 4. From now on, The dependency on the aggregate states is summarized into the t subscript; e.g., $S(\xi, \theta|\Omega_t) = S_t(\xi, \theta)$.

5 Calibration

In the numerical exercise below, I assume two types of workers. I parameterize the overhead cost distribution F with a log-normal distribution of parameters (μ_c, σ_c) and the match-specific productivity distribution H with a log-normal distribution of parameters (μ_ξ, ζ). The parameters in Table 2 are set exogenously.

I use 0.004 for the discount rate, r^C, so that the annual interest rate becomes around 5 percent per year. Standard values are used for the depreciation rate for production capital, $\delta = 0.008$, the decreasing return to capital parameter, $\alpha = 0.3$, and the average monthly UE rate, $p_U(\tau_U) = 0.45$. The disutility of work for the high type is normalized to zero, and the mean type and mean productivity are also normalized to one. I set the exogenous EU rate to 0.02 so that the third tenure group has only exogenous separation

[3] I use automatic differentiation for linearization.

Parameter	Meaning	Value
r^C	Discount rate	0.004
δ	Depreciation rate	0.008
α	DRS parameter with respect to capital input	0.3
$p_U(\tau_U)$	UE rate for unemployed workers	0.45
χ	Exogenous EU rate	0.02
$m(\theta_{high})$	Mass of high surplus type	0.8
$b(\theta_{high})$	Disutility of work for high surplus type	0
μ_c	Mean of log-normal distribution for cost shock	0.1
Mean[θ]	Mean type	1
Mean[ξ]	Mean productivity	1

Table 2: Exogeneously Given Parameters

probability 0.02 in the steady state. I assign arbitrary numbers to other parameters. The parameters in Table 3 are calibrated to match the steady state moments in Table 4, using the simulated method of moments. Note that I also put some penalty on the mean separation probabilities for the high type so that high type workers have negligible probabilities of endogenous separation. In practice, the distribution for ξ is truncated at the first percentile and at the point where the cumulative probability becomes 0.9999. I restrict the equilibrium so that firms with the lowest ξ in the distribution also find it profitable to hire both types of workers.

Parameter	Meaning	Value
Ψ	Steady state frequency of the overhead cost shock	0.59
σ_c	Standard deviation of log overhead cost	0.64
$b(\theta_{low})$	Disutility of work for low surplus type	1.26
ζ	Standard deviation of log match-specific productivity	0.07
η	Firm share of match surplus	0.71
θ_{high}	Productivity of high surplus type	1.01

Table 3: Calibrated Parameters: Steady State

The equilibrium implies the steady state frequency of the overhead cost shock is high; around once in every two months. The standard deviation of the log overhead cost is

large, so the separation probability is a smooth function of match surplus. The disutility of work for the low surplus type is large, around 53 percent of the average wage, while the productivity difference is small. As a result, most of the difference in the separation probability comes from heterogeneity in the disutility of work. The standard deviation of log match-specific productivity is small, because of the equilibrium restriction that even firms with the lowest ξ hire workers. The firm share of the match surplus is larger than the previous literature because workers' outside option is higher than the homogeneous productivity case due to the option value of unemployment. As a result, labor share becomes larger, and the firm share has to increase to match the labor share.

Moment	Value	Target
EU rate	0.024	0.025
EU difference between wage groups(3/1)	0.002	0.002
EU difference between wage groups(2/1)	0.001	0.001
EU difference between tenure groups(3/1)	0.008	0.008
EU difference between tenure groups(2/1)	0.005	0.006
Labor share	0.60	0.60
P90/P10 in wage distribution	1.10	2.80
P50/P10 in wage distribution	1.04	1.7

Table 4: Model and Target Moments: Steady State

The calibrated model can match most of the targeted moments, except for the wage dispersion measure, because of the equilibrium restriction leading to the low value of ξ, and the weight on those moments smaller than others.

Figure 3 has bars representing separation probabilities over wage and tenure bins from the calibrated model. The calibrated model replicates the steady-state pattern on separation rates across wage and tenure groups. Workers with low wage and short tenure are more likely to separate from their employers in a given period. Moreover, the degree of tenure-dependency is larger than the degree of wage-dependency; i.e., tenure is a stronger predictor of EU transition rates than wage.

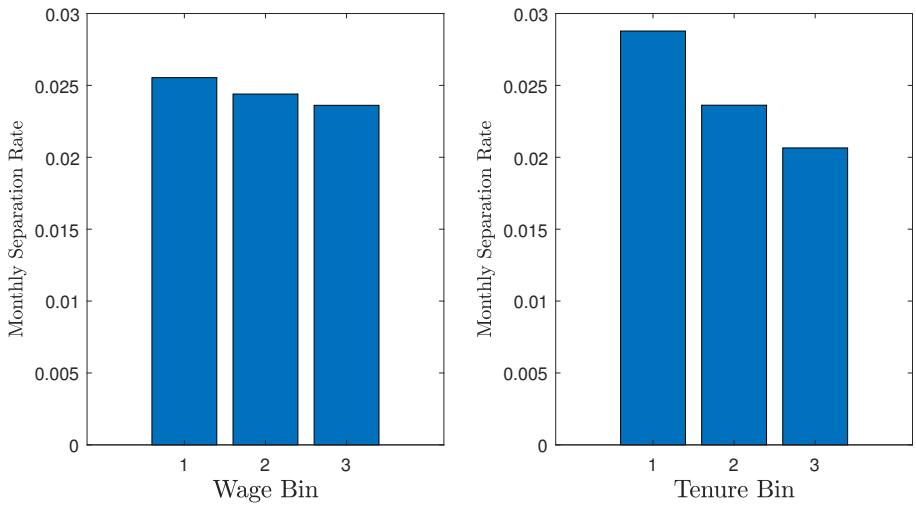


Figure 3: Steady State Separation Probabilities over Wage and Tenure Groups

The left panel of figure 4 expresses the density of workers over the match surplus across the two types. The low surplus type has smaller dispersion since the lower part of the distribution exits from the employment at faster rates. Because of the selection of workers over time, the probability that a worker is a low surplus type decreases fast with the tenure, especially when a worker has a short tenure.

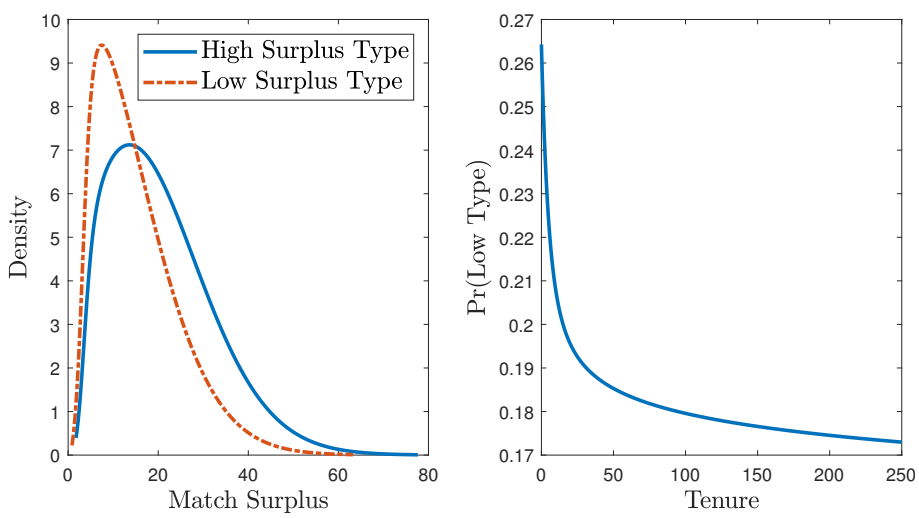


Figure 4: Steady State Distribution of Workers

Parameters in the stochastic process for Ψ and the match elasticity, γ, also appear in the

log-linearized system of equations characterizing the economy with aggregate uncertainty. Note that the output Y_t is defined as the sum of individual output *before* paying the overhead cost. Table 5 and 6 show the parameters used for numerical exercises and the targeted values.

Parameter	Meaning	Value
β	Mean reversion parameter for Ψ process	0.118
σ	Volatility parameter for Ψ process	0.208
γ	Match elasticity	0.488

Table 5: Calibrated Parameters: Business Cycle

Moment	Value	Target
Std($\log(Y_t)$)	0.04	0.04
Std(UE_t)	0.05	0.05
Std(EU_t)	0.005	0.004
Autocorr(UE_t, UE_{t-3})	0.80	0.80

Table 6: Model and Target Moments: Business Cycle

6 Aggregate Implications

6.1 Impulse Response Functions

Figure 5 displays the impulse responses of labor market flow rates to one standard deviation Dirac-Delta shock to Ψ_t at $t = 0$, which decays with constant rate afterwards. An increase in the frequency of overhead cost shocks reduces the match surplus for all matches. Consequently, the job-finding rate decreases from 4.5 to 4.32 and recovers slowly. The separation rate jumps up by 0.5 percent point per month and returns to the steady state level at a quicker speed.

Responses of aggregate variables are shown in figure 6. In response to a shock raising the frequency of overhead cost shock by 0.2, the unemployment rate reaches 5.7 percent, 0.5 percent higher than the steady state level. The decrease in employment reduces output

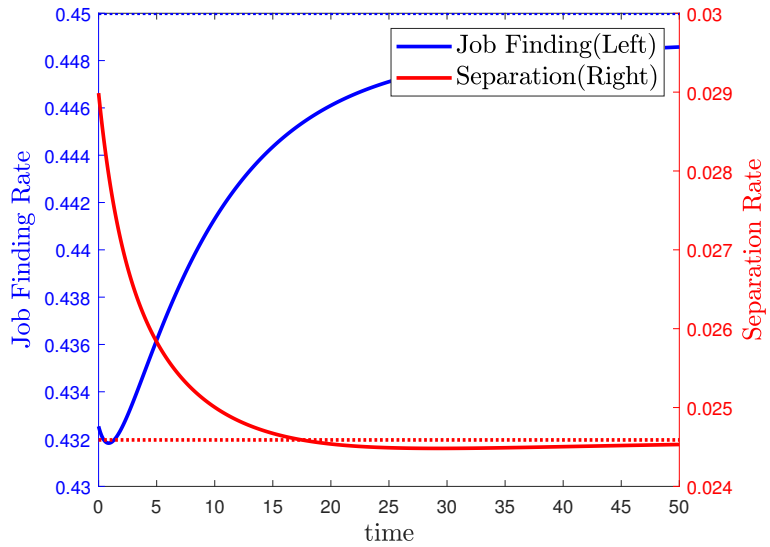


Figure 5: Impulse Responses Functions: job-finding and Separation Rates

initially, and long-run decrease in output comes from a decrease in production capital and interest rate higher than the steady state level.

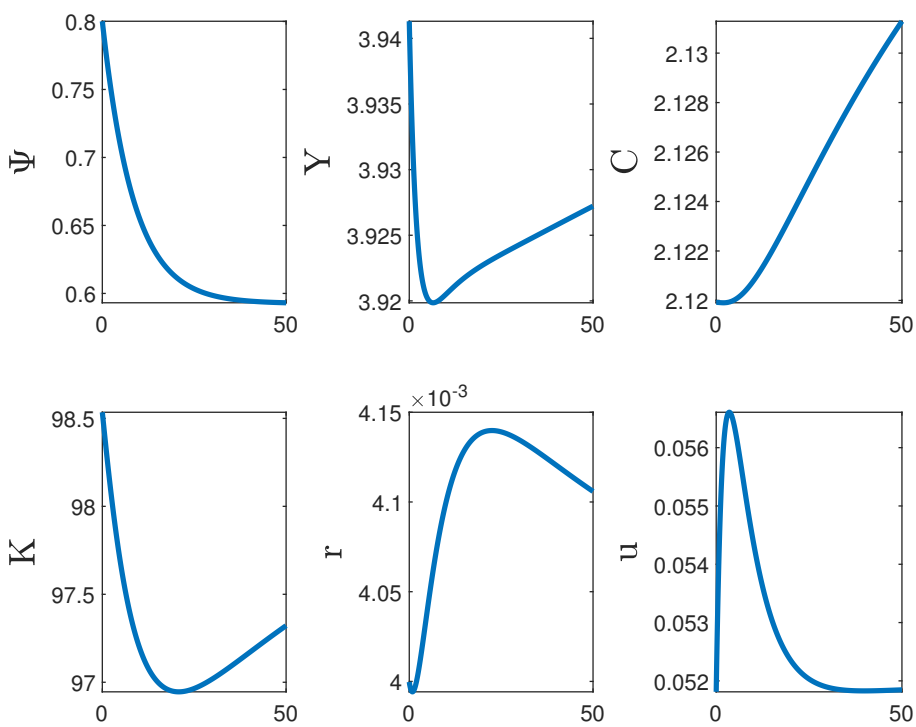


Figure 6: Impulse Response Functions: Aggregate Variables

6.2 Increase in Separation Probabilities over Wage and Tenure Groups

Figure 7 shows increases in separation rates over different wage and tenure bins, instantaneously after the Dirac-Delta shock. The calibrated model gives patterns qualitatively similar to the ones in the CPS data; the last wage bin and the first tenure bin experience the largest increases in separation rates.

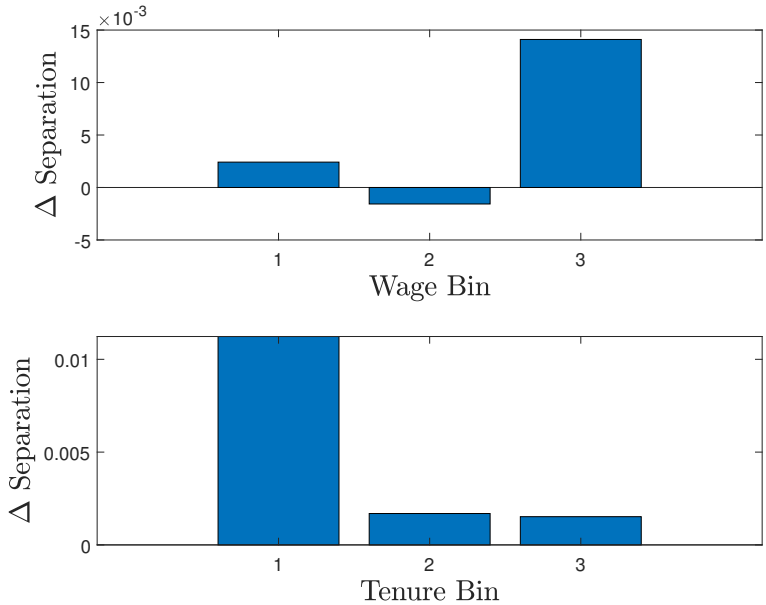


Figure 7: Increase in Separation Rates over Wage and Tenure Bins

A component that raises the wage but *reduces* the match surplus, disutility of work in this model, is essential to break the one-to-one correspondence between tenure (or persistently low separation rate) and wage. Low type workers with low match-specific productivity have higher wage because their disutility of work is higher, not because they are more productive. They are more likely to separate into unemployment because of this higher disutility of work. Having a short job tenure signals the type of worker through selection, so the tenure is a strong predictor of separation probabilities.

The cyclical shock in Ψ_t, frequency of overhead cost shock, hits the low type workers

with low match-specific productivity hardest, as the density of the overhead cost shock is higher around the value of the match with these workers. These workers have high wage and short job tenure, replicating the empirical data pattern. These workers can be thought of as the lemons that survived the screening process but gives a small profit to the employer.

6.3 Sorting and Persistence of Labor Market Variables

Figure 8 expresses the change in the composition of unemployed workers after the Dirac-Delta shock. The fraction of workers that are low type increases from 26.5 to 28 percent and stays higher than the steady-state level for a while. The share of low type in unemployed workers is modest in magnitude because separation probabilities jump up but recover quickly to the pre-recession level. Only a small fraction of matches with low type workers are affected by the cyclical shock.

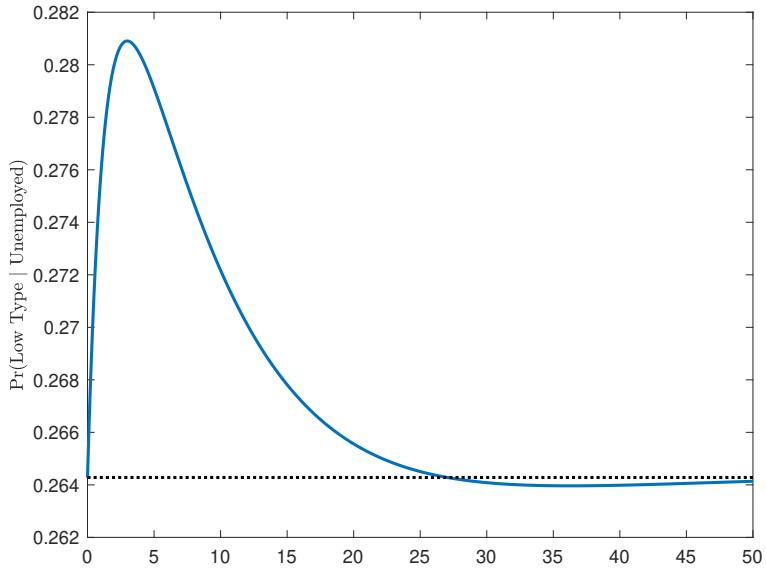


Figure 8: Fraction of Unemployment Workers that are Low Type

The selection of low type into unemployment implies the composition effect works to decrease the entrants' value from job creation in recession. To see the qualitative and

quantitative implications of worker sorting over the business cycle, I compare the impulse responses of job-finding and unemployment rates from the benchmark model to the ones from a counterfactual scenario without worker composition effect in job creation decision. I substitute the free entry condition of the benchmark model with the following equation.

$$c = q(\tau_t) \int \sum_{\theta} \frac{u(\theta)}{u} \eta S_t(\theta, \xi) dH(\xi).$$

The only difference is that the composition term, $u(\theta)/u$, uses the steady state value instead of the actual one $u_t(\theta)/u_t$. Note that this is a partial equilibrium analysis, in the sense that the interest rate r_t as well as the value $S_t(\theta, \xi)$ determining separation rates between workers and employers stay the same as the benchmark case with sorting, and only the free entry condition determining labor market tightness changes.

Figure 9 compares the responses of job-finding probabilities and the unemployment rates in the benchmark case and to the case without worker sorting.

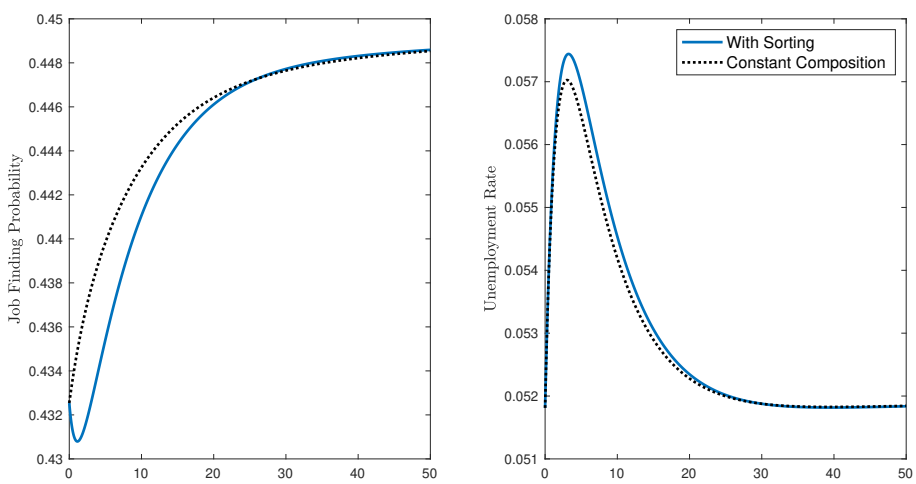


Figure 9: Impulse Responses with and without Worker Sorting

Sorting contributes to hump-shaped responses of job-finding rates since the composition among unemployed workers is a slow-moving state variable. job-finding probability fluc-

tuates more persistently, and the unemployment rate increases by more. However, the impact of sorting is modest quantitatively, especially on unemployment rates.

The reason for the small quantitative impact is the failure of the model in generating sufficient influx to unemployment through separation. Most of the initial increase in the unemployment rate is due to the sharp decrease in job-finding probabilities. The model assumes no selection for the hiring process, so the small separation translates into a small impact of sorting. Data suggests that a higher amount of separation contributes to a substantial decrease in employment at the onset of the recession.

The impact of sorting can be magnified if the considered scenario is such that the Ψ stays high for a while, then goes back to the steady state level, instead of the rapidly decaying Dirac-Delta shock. In this case, the separation may also play a large role in increasing the unemployment rate, with a higher amount of low type workers into the unemployment pool.

Another way to amplify the role of sorting is to incorporate the directed search in the hiring process. If only some of the unemployed workers are mixed with the separated workers, due to the signaling effect of unemployment duration, for instance, even small amount of endogenous separation applied to low type workers may change the entry value through composition effect by a larger amount.

7 Concluding Remarks

I interpret the separation as a selective process for firms to sort low type workers into unemployment. Data suggests the selection is stronger in recessions, as implied by increases in separation rates concentrated around workers with short tenure. Short tenure signals low surplus type workers in endogeneous separation models. In potential entrants' perspective, stronger sorting in recessions means a lower value of creating new jobs, as they are more likely to end up hiring low type workers. In spite of modest magnitude, sorting amplifies the impact of aggregate productivity shocks in the labor market through

larger fluctuations in job-finding rates. This mechanism can provide recent literature on regional variation of employment dynamics with explanations on why workers exposed to regional labor market shocks during the Great Recession still have lower employment rates even in the mid-2010s, as pointed out by Yagan (2019), and why massive local layoffs have larger impacts on workers unemployed at the timing of the event in Germany, as in Gathmann et al. (2017).

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A Appendix

A.1 Derivation of Nash Bargained Wage

Wage is set to maximize the following object.

$$w = \arg\max_{w} \left(\bar{E}(\theta, \xi, w|\Omega) - U(\theta|\Omega)\right)^{1-\eta} \left(\bar{V}(\theta, \xi, w|\Omega)\right)^{\eta}$$

Recall that the two agents decide the amount of overhead cost paid by each through Nash bargaining with the same share parameter.

$$X(c) = (1-\eta)c$$
$$c - X(c) = \eta c$$

Let's define $\bar{S}(\theta, \xi, w|\Omega) = \bar{V}(\theta, \xi, w|\Omega) + \bar{E}(\theta, \xi, w|\Omega) - U(\theta|\Omega)$. Then,

$$\begin{aligned} r\bar{S}(\theta, \xi, w|\Omega) = \pi(\xi|\Omega) &+ \left(\chi + \Psi\mathbb{P}[\bar{S}(\theta, \xi, w|\Omega) < c]\right)\left(-\bar{S}(\theta, \xi, w|\Omega)\right) \\ &- b(\theta) - p(\tau)\int \left(E(\theta, \tilde{\xi}|\Omega) - U(\theta|\Omega)\right) dH(\tilde{\xi}) \\ &- \Psi\mathbb{E}[c|\bar{S}(\theta, \xi, w|\Omega) \geq c]\mathbb{P}[\bar{S}(\theta, \xi, w|\Omega) \geq c] + \frac{d}{dt}\mathbb{E}[S(\theta, \xi|\Omega)|\Omega] \end{aligned}$$

This gives

$$\frac{d\bar{S}(\theta, \xi, w|\Omega)}{dw} = 0,$$

and $\bar{S}(\theta, \xi, w|\Omega) = S(\theta, \xi|\Omega)$. Moreover,

$$\begin{aligned} \left(r + \chi + \Psi\mathbb{P}\left[S(\theta, \xi|\Omega) < c\right]\right)\frac{d\bar{E}(\theta, \xi, w|\Omega)}{dw} &= 1 + \Psi\eta f(S(\theta, \xi|\Omega))S(\theta, \xi|\Omega)\frac{dS(\theta, \xi|\Omega)}{dw} \\ &= 1 \end{aligned}$$

Likewise,

$$\Big(r+\chi+\Psi\mathbb{P}\left[S(\theta,\xi|\Omega)<c\right]\Big)\frac{d\bar{V}(\theta,\xi,w|\Omega)}{dw}=-1$$

The first order condition of the wage setting problem is

$$\begin{aligned}(1-\eta)\left(\bar{V}(\theta,\xi,w|\Omega)\right)&=\eta\left(\bar{E}(\theta,\xi,w|\Omega)-U(\theta|\Omega)\right)\\
(1-\eta)\Bigg\{\pi(\xi|\Omega)-w-\Psi\eta\mathbb{C}(S(\theta,\xi|\Omega))&=\eta\Bigg\{w-b(\theta)-p(\tau)\int\Big(E(\theta,\tilde{\xi}|\Omega)-U(\theta|\Omega)\Big)dH(\tilde{\xi})\\
&\quad+\frac{d}{dt}\mathbb{E}[\bar{V}(\theta,\xi,w|\Omega)|\Omega]\Bigg\}\quad-\Psi(1-\eta)\mathbb{C}(S(\theta,\xi|\Omega))+\frac{d}{dt}\mathbb{E}[\bar{E}(\theta,\xi,w|\Omega)-U(\theta|\Omega)|\Omega]\Bigg\}\end{aligned}$$

where $\mathbb{C}(S(\theta,\xi|\Omega))=\mathbb{E}[c|c\leq S(\theta,\xi|\Omega)]\mathbb{P}[c\leq S(\theta,\xi|\Omega)]$.
Rearranging the equation above, using $(1-\eta)\frac{d}{dt}\mathbb{E}[\bar{V}(\theta,\xi,w|\Omega)|\Omega]=\eta\frac{d}{dt}\mathbb{E}[\bar{E}(\theta,\xi,w|\Omega)-U(\theta|\Omega)|\Omega]$

$$\begin{aligned}w(\theta,\xi)&=(1-\eta)\pi(\xi|\Omega)-\eta b(\theta)+\eta p(\tau)\int\Big(E(\theta,\tilde{\xi}|\Omega)-U(\theta|\Omega)\Big)dH(\tilde{\xi})\\
&=(1-\eta)\pi(\xi|\Omega)-\eta b(\theta)+\eta(1-\eta)p(\tau)\int S(\xi,\theta|\Omega)dH(\tilde{\xi})\end{aligned}$$

This expression gives

$$\begin{aligned}\big(r+\chi+\delta(S(\theta,\xi|\Omega))\big)\left(E(\theta,\xi|\Omega)-U(\theta|\Omega)\right)&=(1-\eta)\Bigg(\pi(\xi|\Omega)\\
&\quad-b(\theta)-p(\tau)\int\Big(E(\theta,\tilde{\xi}|\Omega)-U(\theta|\Omega)\Big)dH(\tilde{\xi})\\
&\quad-\Psi\mathbb{C}(S(\theta,\xi|\Omega))+\frac{d}{dt}\mathbb{E}[E(\theta,\xi|\Omega)-U(\theta|\Omega)|\Omega]\Bigg)\\
\big(r+\chi+\delta(S(\xi,\theta|\Omega))\big)V(\theta,\xi|\Omega)&=\eta\Bigg(\pi(\xi|\Omega)-b(\theta)-p(\tau)\int\Big(E(\theta,\tilde{\xi}|\Omega)-U(\theta|\Omega)\Big)dH(\tilde{\xi})\\
&\quad-\Psi\mathbb{C}(S(\theta,\xi|\Omega))+\frac{d}{dt}\mathbb{E}[V(\theta,\xi)|\Omega]\Bigg)\end{aligned}$$

Then,

$$E(\theta, \xi|\Omega) - U(\theta|\Omega) = (1-\eta)S(\theta, \xi|\Omega)$$
$$V(\theta, \xi|\Omega) = \eta S(\theta, \xi|\Omega)$$

where again

$$\begin{aligned} rS(\theta, \xi|\Omega) = \pi(\xi|\Omega) - b(\theta) - p(\tau)\int (1-\eta)S(\theta, \tilde{\xi}|\Omega)dH(\tilde{\xi}) \\ - \Psi\mathbb{C}(S(\theta, \xi|\Omega)) - (\chi + \delta(S(\theta, \xi|\Omega)))S(\theta, \xi|\Omega) + \frac{d}{dt}\mathbb{E}[S(\theta, \xi|\Omega)|\Omega] \end{aligned}$$

A.2 Full Characterization of the Economy with Aggregate Uncertainty

Recall again that, for notational simplicity, I summarize the dependency on the aggregate states to t subscript; i.e. $S(\xi, \theta|\Omega_t) = S_t(\xi, \theta)$, and other variables are denoted similarly.

$$
\begin{aligned}
\frac{d}{dt} g_t(\theta, \xi) &= p(\tau_t) u_t(\theta) h(\xi) - (\chi + \delta_t(\theta, \xi)) g_t(\theta, \xi) \\
\frac{d}{dt} K_t + c_t + c_v v_t &= \left(\frac{\alpha}{r_t}\right)^{\frac{\alpha}{1-\alpha}} \sum_\theta \int (\xi\theta)^{\frac{1}{1-\alpha}} g_t(\theta, \xi) d\xi - \sum_\theta \Psi_t \int \mathbb{C}(S_t(\theta, \xi)) g_t(\xi, \theta) d\xi - \delta K_t \\
K_t &= \left(\frac{\alpha}{r_t}\right)^{\frac{1}{1-\alpha}} \sum_\theta \int (\xi\theta)^{\frac{1}{1-\alpha}} g_t(\theta, \xi) d\xi \\
d(\Psi_t - \Psi) &= -\beta(\Psi_t - \Psi) dt + \sigma dW_t \\
\frac{d}{dt} \mathbb{E}[S_t(\theta, \xi)] &= (r_t + \chi + \delta_t(\theta, \xi)) S_t(\theta, \xi) - \pi_t(\theta, \xi) - b(\theta) - \Psi_t \mathbb{C}(S_t(\theta, \xi)) \\
&\quad - p(\tau_t) \eta (1 - \eta) \int S_t(\theta, \tilde{\xi}) dH(\tilde{\xi}) \\
\delta_t(\theta, \xi) &= \Psi_t (1 - F(S_t(\theta, \xi))) \\
\frac{d}{dt} \mathbb{E}[c_t^{-1}] &= (\rho + \delta - r_t) c_t^{-1} \\
c &= q(\tau_t) \int \sum_\theta \frac{u_t(\theta)}{u_t} \eta S_t(\theta, \xi) dH(\xi)
\end{aligned}
$$